Modul 03 Basis Data

PRAKTIKUM BASIS DATA

Scripting Sql Basic Query

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SOAL TP 1

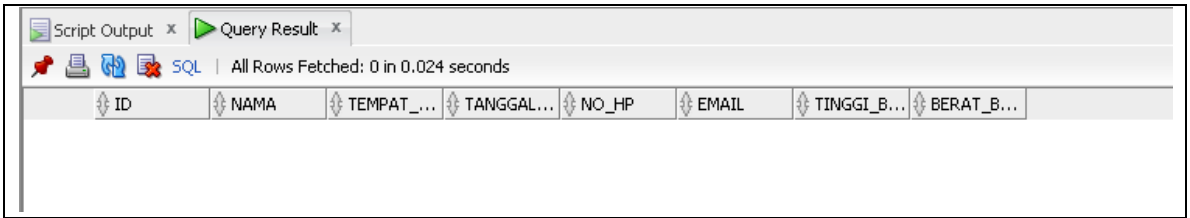
Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan

JAWAB

KODE

```
CREATE TABLE mahasiswa (
   id NUMBER PRIMARY KEY,
   nama VARCHAR2(100),
   tempat_lahir VARCHAR2(50),
   tanggal_lahir DATE,
   no_hp VARCHAR2(15),
   email VARCHAR2(100),
   tinggi_badan NUMBER,
   berat_badan NUMBER
);
```

OUTPUT



SOAL TP 2

Lakukan insert data pada table nomor 1 sebanyak 15 rows.

JAWAB

KODE

```
INSERT INTO mahasiswa VALUES (1,'Andi','Jakarta',DATE '2002-01-
10','081234567890','andi@gmail.com',170,60);
INSERT INTO mahasiswa VALUES (2,'Budi','Bandung',DATE '2001-02-
12','081234567891','budi@gmail.com',168,58);
INSERT INTO mahasiswa VALUES (3,'Citra','Surabaya',DATE '2002-03-
15','081234567892','citra@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (4,'Dina','Medan',DATE '2001-04-
20','081234567893','dina@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (5,'Eko','Semarang',DATE '2002-05-
05','081234567894','eko@gmail.com',175,65);
INSERT INTO mahasiswa VALUES (6,'Fajar','Bogor',DATE '2001-06-
18','081234567895','fajar@gmail.com',172,63);
INSERT INTO mahasiswa VALUES (7,'Gina','Depok',DATE '2002-07-
22','081234567896','gina@gmail.com',158,48);
INSERT INTO mahasiswa VALUES (8,'Hadi','Bekasi',DATE '2001-08-
30','081234567897','hadi@gmail.com',180,70);
INSERT INTO mahasiswa VALUES (9,'Intan','Padang',DATE '2002-09-
10','081234567898','intan@gmail.com',165,55);
INSERT INTO mahasiswa VALUES (10,'Joko','Malang',DATE '2001-10-
11','081234567899','joko@gmail.com',173,66);
INSERT INTO mahasiswa VALUES (11,'Kiki','Bali',DATE '2002-11-
12','081234567800','kiki@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (12,'Lina','Yogyakarta',DATE '2001-12-
13','081234567801','lina@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (13,'Mira','Solo',DATE '2002-01-
14','081234567802','mira@gmail.com',159,49);
INSERT INTO mahasiswa VALUES (14,'Nanda','Lampung',DATE '2001-02-
15','081234567803','nanda@gmail.com',174,67);
INSERT INTO mahasiswa VALUES (15,'Oki','Palembang',DATE '2002-03-
16','081234567804','oki@gmail.com',171,64);
```

OUTPUT

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.013 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	15	Oki	Palembang	16-MAR-02	081234567804	oki@gmail.com	171	64
2	14	Nanda	Lampung	15-FEB-01	081234567803	nanda@gmail.com	174	67
3	13	Mira	Solo	14-JAN-02	081234567802	mira@gmail.com	159	49
4	1	Andi	Jakarta	10-JAN-02	081234567890	andi@gmail.com	170	60
5	2	Budi	Bandung	12-FEB-01	081234567891	budi@gmail.com	168	58
6	3	Citra	Surabaya	15-MAR-02	081234567892	citra@gmail.com	160	50
7	4	Dina	Medan	20-APR-01	081234567893	dina@gmail.com	162	52
8	5	Eko	Semarang	05-MAY-02	081234567894	eko@gmail.com	175	65
9	6	Fajar	Bogor	18-JUN-01	081234567895	fajar@gmail.com	172	63
10	7	Gina	Depok	22-JUL-02	081234567896	gina@gmail.com	158	48
11	8	Hadi	Bekasi	30-AUG-01	081234567897	hadi@gmail.com	180	70
12	9	Intan	Padang	10-SEP-02	081234567898	intan@gmail.com	165	55
13	10	Joko	Malang	11-OCT-01	081234567899	joko@gmail.com	173	66
14	11	Kiki	Bali	12-NOV-02	081234567800	kiki@gmail.com	160	50
15	12	Lina	Yogyakarta	13-DEC-01	081234567801	lina@gmail.com	162	52

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:

a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)

```
SELECT LOWER(nama) FROM mahasiswa;

SELECT UPPER(nama) FROM mahasiswa;

SELECT INITCAP(nama) FROM mahasiswa;

SELECT CONCAT(nama, ' - Mahasiswa') FROM mahasiswa;

SELECT nama, LENGTH(nama) FROM mahasiswa;

SELECT SUBSTR(nama,1,3) FROM mahasiswa;

SELECT LPAD(nama,10,'*') FROM mahasiswa;

SELECT TRIM('  ANDI  ') FROM dual;

SELECT INSTR(nama,'a') FROM mahasiswa;

SELECT REPLACE(nama,'a','@') FROM mahasiswa;
```

OUTPUT

Script Output | Query Result

All Rows Fetched: 15 in 0.032 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN	NAMA_LOWER	NAMA_UPPER	NAMA_INITCAP	NAMA_CONCAT
1	15	Oki	Palembang	16-MAR-02	081234567804	oki@gmail.com	171	64	oki	OKI	Oki	Oki_Mahasiswa
2	14	Nanda	Lampung	15-FEB-01	081234567803	nanda@gmail.com	174	67	nanda	NANDA	Nanda	Nanda_Mahasis...
3	13	Mira	Solo	14-JAN-02	081234567802	mira@gmail.com	159	49	mira	MIRA	Mira	Mira_Mahasiswa
4	1	Andi	Jakarta	10-JAN-02	081234567890	andi@gmail.com	170	60	andi	ANDI	Andi	Andi_Mahasiswa
5	2	Budi	Bandung	12-FEB-01	081234567891	budi@gmail.com	168	58	budi	BUDI	Budi	Budi_Mahasiswa
6	3	Citra	Surabaya	15-MAR-02	081234567892	citra@gmail.com	160	50	citra	CITRA	Citra	Citra_Mahasiswa
7	4	Dina	Medan	20-APR-01	081234567893	dina@gmail.com	162	52	dina	DINA	Dina	Dina_Mahasiswa
8	5	Eko	Semarang	05-MAY-02	081234567894	eko@gmail.com	175	65	eko	EKO	Eko	Eko_Mahasiswa
9	6	Fajar	Bogor	18-JUN-01	081234567895	fajar@gmail.com	172	63	fajar	FAJAR	Fajar	Fajar_Mahasiswa
10	7	Gina	Depok	22-JUL-02	081234567896	gina@gmail.com	158	48	gina	GINA	Gina	Gina_Mahasiswa
11	8	Hadi	Bekasi	30-AUG-01	081234567897	hadi@gmail.com	180	70	hadi	HADI	Hadi	Hadi_Mahasiswa
12	9	Intan	Padang	10-SEP-02	081234567898	intan@gmail.com	165	55	intan	INTAN	Intan	Intan_Mahasiswa
13	10	Joko	Malang	11-OCT-01	081234567899	joko@gmail.com	173	66	joko	JOKO	Joko	Joko_Mahasiswa
14	11	Kiki	Bali	12-NOV-02	081234567800	kiki@gmail.com	160	50	kiki	KIKI	Kiki	Kiki_Mahasiswa
15	12	Lina	Yogyakarta	13-DEC-01	081234567801	lina@gmail.com	162	52	lina	LINA	Lina	Lina_Mahasiswa

PANJANG_NAMA	POTONGAN_NAMA	LPAD_NAMA	RPAD_NAMA	LTRIM_NAMA	RTRIM_NAMA	POSISI_A	REPLACE_NAMA
3	Oki	*******Oki	Oki*******	Oki	Oki	0	Oki
5	Nan	*****Nanda	Nanda*****	Nanda	Nanda	2	N@nd@
4	Mir	******Mira	Mira******	Mira	Mira	4	Mir@
4	And	******Andi	Andi******	Andi	Andi	0	Andi
4	Bud	******Budi	Budi******	Budi	Budi	0	Budi
5	Cit	*****Citra	Citra*****	Citra	Citra	5	Citr@
4	Din	******Dina	Dina******	Dina	Dina	4	Din@
3	Eko	*******Eko	Eko*******	Eko	Eko	0	Eko
5	Faj	*****Fajar	Fajar*****	Fajar	Fajar	2	F@j@r
4	Gin	******Gina	Gina******	Gina	Gina	4	Gin@
4	Had	******Hadi	Hadi******	Hadi	Hadi	2	H@di
5	Int	*****Intan	Intan*****	Intan	Intan	4	Int@n
4	Jok	******Joko	Joko******	Joko	Joko	0	Joko
4	Kik	******Kiki	Kiki******	Kiki	Kiki	0	Kiki
4	Lin	******Lina	Lina******	Lina	Lina	4	Lin@

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH|YEAR]))

```
SELECT SYSDATE FROM dual;

SELECT ADD_MONTHS(SYSDATE,3) FROM dual;

SELECT LAST_DAY(SYSDATE) FROM dual;

SELECT NEXT_DAY(SYSDATE,'MONDAY') FROM dual;

SELECT MONTHS_BETWEEN(SYSDATE, tanggal_lahir) FROM mahasiswa;

SELECT ROUND(SYSDATE,'MONTH') FROM dual;
```

```
SELECT ROUND(SYSDATE,'YEAR') FROM dual;
```

OUTPUT

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.037 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN	BULAN_ANGKA	BULAN_SINGKAT	BULAN_LENGKAP	HARI_TANGGAL
1	15	Oki	Palembang	16-MAR-02	081234567804	oki@gmail.com	171	64 03	MAR	MARCH	16
2	14	Nanda	Lampung	15-FEB-01	081234567803	nanda@gmail.com	174	67 02	FEB	FEBRUARY	15
3	13	Mira	Solo	14-JAN-02	081234567802	mira@gmail.com	159	49 01	JAN	JANUARY	14
4	1	Andi	Jakarta	10-JAN-02	081234567890	andi@gmail.com	170	60 01	JAN	JANUARY	10
5	2	Budi	Bandung	12-FEB-01	081234567891	budi@gmail.com	168	58 02	FEB	FEBRUARY	12
6	3	Citra	Surabaya	15-MAR-02	081234567892	citra@gmail.com	160	50 03	MAR	MARCH	15
7	4	Dina	Medan	20-APR-01	081234567893	dina@gmail.com	162	52 04	APR	APRIL	20
8	5	Eko	Semarang	05-MAY-02	081234567894	eko@gmail.com	175	65 05	MAY	MAY	05
9	6	Fajar	Bogor	18-JUN-01	081234567895	fajar@gmail.com	172	63 06	JUN	JUNE	18
10	7	Gina	Depok	22-JUL-02	081234567896	gina@gmail.com	158	48 07	JUL	JULY	22
11	8	Hadi	Bekasi	30-AUG-01	081234567897	hadi@gmail.com	180	70 08	AUG	AUGUST	30
12	9	Intan	Padang	10-SEP-02	081234567898	intan@gmail.com	165	55 09	SEP	SEPTEMBER	10
13	10	Joko	Malang	11-OCT-01	081234567899	joko@gmail.com	173	66 10	OCT	OCTOBER	11
14	11	Kiki	Bali	12-NOV-02	081234567800	kiki@gmail.com	160	50 11	NOV	NOVEMBER	12
15	12	Lina	Yogyakarta	13-DEC-01	081234567801	lina@gmail.com	162	52 12	DEC	DECEMBER	13

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.037 seconds

	L	HARI_SINGKAT	TAHUN_LENGKAP	TAHUN_SINGKAT	TAHUN_RR	AM_PM	JAM_12	JAM_24	MENIT	DETIK	TAMBAH_3_BULAN	TANGGAL_SEKARANG	AKHIR_BULAN
1		SAT	2002	02	02	AM	12	00	00	22	16-JUN-02	06-MAR-26	31-MAR-02
2		THU	2001	01	01	AM	12	00	00	22	15-MAY-01	06-MAR-26	28-FEB-01
3		MON	2002	02	02	AM	12	00	00	22	14-APR-02	06-MAR-26	31-JAN-02
4		THU	2002	02	02	AM	12	00	00	22	10-APR-02	06-MAR-26	31-JAN-02
5		MON	2001	01	01	AM	12	00	00	22	12-MAY-01	06-MAR-26	28-FEB-01
6		FRI	2002	02	02	AM	12	00	00	22	15-JUN-02	06-MAR-26	31-MAR-02
7		FRI	2001	01	01	AM	12	00	00	22	20-JUL-01	06-MAR-26	30-APR-01
8		SUN	2002	02	02	AM	12	00	00	22	05-AUG-02	06-MAR-26	31-MAY-02
9		MON	2001	01	01	AM	12	00	00	22	18-SEP-01	06-MAR-26	30-JUN-01
10		MON	2002	02	02	AM	12	00	00	22	22-OCT-02	06-MAR-26	31-JUL-02
11		THU	2001	01	01	AM	12	00	00	22	30-NOV-01	06-MAR-26	31-AUG-01
12		TUE	2002	02	02	AM	12	00	00	22	10-DEC-02	06-MAR-26	30-SEP-02
13		THU	2001	01	01	AM	12	00	00	22	11-JAN-02	06-MAR-26	31-OCT-01
14		TUE	2002	02	02	AM	12	00	00	22	12-FEB-03	06-MAR-26	30-NOV-02
15		THU	2001	01	01	AM	12	00	00	22	13-MAR-02	06-MAR-26	31-DEC-01

SENIN_BERIKUTNYA	SELISIH_BULAN	ROUND_BULAN	ROUND_TAHUN
18-MAR-02	287.67742756869772998805256869772998805 3	01-APR-02	01-JAN-02
19-FEB-01	300.70968563321385902031063321385902031 1	01-FEB-01	01-JAN-01
21-JAN-02	289.74194369772998805256869772998805256 9	01-JAN-02	01-JAN-02
14-JAN-02	289.87097595579450418160095579450418160 1	01-JAN-02	01-JAN-02
19-FEB-01	300.80645982676224611708482676224611708 5	01-FEB-01	01-JAN-01
18-MAR-02	287.70968563321385902031063321385902031 1	01-MAR-02	01-JAN-02
23-APR-01	298.54839531063321385902031063321385902	01-MAY-01	01-JAN-01
06-MAY-02	286.03226627837514934289127837514934289 1	01-MAY-02	01-JAN-02
25-JUN-01	296.61291143966547192353643966547192353 6	01-JUL-01	01-JAN-01
29-JUL-02	283.48387918160095579450418160095579450 4	01-AUG-02	01-JAN-03
03-SEP-01	294.22581466547192353643966547192353644	01-SEP-01	01-JAN-02
16-SEP-02	281.87097595579450418160095579450418160 1	01-SEP-02	01-JAN-03
15-OCT-01	292.83871789127837514934289127837514934 3	01-OCT-01	01-JAN-02
18-NOV-02	279.80645982676224611708482676224611708 5	01-NOV-02	01-JAN-03
17-DEC-01	290.77420176224611708482676224611708482 7	01-DEC-01	01-JAN-02

C. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

```
SELECT TO_NUMBER('123') FROM dual;

SELECT TO_CHAR(SYSDATE,'DD-MM-YYYY') FROM dual;

SELECT TO_CHAR(12345) FROM dual;

SELECT TO_DATE('10-01-2020','DD-MM-YYYY') FROM dual;
```

OUTPUT

Script Output | Query Result

All Rows Fetched: 15 in 0.016 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN	CONTOH_TO_NUMBER	TANGGAL_CHAR	TINGGI_CHAR	CONTOH_TO_I
1	15	Oki	Palembang	16-MAR-02	081234567804	oki@gmail.com	171	64	123	16-03-2002	171	10-JAN-20
2	14	Nanda	Lampung	15-FEB-01	081234567803	nanda@gmail.com	174	67	123	15-02-2001	174	10-JAN-20
3	13	Mira	Solo	14-JAN-02	081234567802	mira@gmail.com	159	49	123	14-01-2002	159	10-JAN-20
4	1	Andi	Jakarta	10-JAN-02	081234567890	andi@gmail.com	170	60	123	10-01-2002	170	10-JAN-20
5	2	Budi	Bandung	12-FEB-01	081234567891	budi@gmail.com	168	58	123	12-02-2001	168	10-JAN-20
6	3	Citra	Surabaya	15-MAR-02	081234567892	citra@gmail.com	160	50	123	15-03-2002	160	10-JAN-20
7	4	Dina	Medan	20-APR-01	081234567893	dina@gmail.com	162	52	123	20-04-2001	162	10-JAN-20
8	5	Eko	Semarang	05-MAY-02	081234567894	eko@gmail.com	175	65	123	05-05-2002	175	10-JAN-20
9	6	Fajar	Bogor	18-JUN-01	081234567895	fajar@gmail.com	172	63	123	18-06-2001	172	10-JAN-20
10	7	Gina	Depok	22-JUL-02	081234567896	gina@gmail.com	158	48	123	22-07-2002	158	10-JAN-20
11	8	Hadi	Bekasi	30-AUG-01	081234567897	hadi@gmail.com	180	70	123	30-08-2001	180	10-JAN-20
12	9	Intan	Padang	10-SEP-02	081234567898	intan@gmail.com	165	55	123	10-09-2002	165	10-JAN-20
13	10	Joko	Malang	11-OCT-01	081234567899	joko@gmail.com	173	66	123	11-10-2001	173	10-JAN-20
14	11	Kiki	Bali	12-NOV-02	081234567800	kiki@gmail.com	160	50	123	12-11-2002	160	10-JAN-20
15	12	Lina	Yogyakarta	13-DEC-01	081234567801	lina@gmail.com	162	52	123	13-12-2001	162	10-JAN-20

D. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

```
SELECT ABS(-10) FROM dual;

SELECT MOD(10,3) FROM dual;

SELECT FLOOR(12.7) FROM dual;

SELECT POWER(2,3) FROM dual;

SELECT ROUND(12.567,2) FROM dual;

SELECT CEIL(12.1) FROM dual;
```

OUTPUT

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.016 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN	CONTOH_ABS	CONTOH_MOD	CONTOH_FLOOR	CONTOH_POWER
1	15	Oki	Palembang	16-MAR-02	081234567804	oki@gmail.com	171	64	10	0	64	8
2	14	Nanda	Lampung	15-FEB-01	081234567803	nanda@gmail.com	174	67	10	0	67	8
3	13	Mira	Solo	14-JAN-02	081234567802	mira@gmail.com	159	49	10	0	49	8
4	1	Andi	Jakarta	10-JAN-02	081234567890	andi@gmail.com	170	60	10	2	60	8
5	2	Budi	Bandung	12-FEB-01	081234567891	budi@gmail.com	168	58	10	0	58	8
6	3	Citra	Surabaya	15-MAR-02	081234567892	citra@gmail.com	160	50	10	1	50	8
7	4	Dina	Medan	20-APR-01	081234567893	dina@gmail.com	162	52	10	0	52	8
8	5	Eko	Semarang	05-MAY-02	081234567894	eko@gmail.com	175	65	10	1	65	8
9	6	Fajar	Bogor	18-JUN-01	081234567895	fajar@gmail.com	172	63	10	1	63	8
10	7	Gina	Depok	22-JUL-02	081234567896	gina@gmail.com	158	48	10	2	48	8
11	8	Hadi	Bekasi	30-AUG-01	081234567897	hadi@gmail.com	180	70	10	0	70	8
12	9	Intan	Padang	10-SEP-02	081234567898	intan@gmail.com	165	55	10	0	55	8
13	10	Joko	Malang	11-OCT-01	081234567899	joko@gmail.com	173	66	10	2	66	8
14	11	Kiki	Bali	12-NOV-02	081234567800	kiki@gmail.com	160	50	10	1	50	8
15	12	Lina	Yogyakarta	13-DEC-01	081234567801	lina@gmail.com	162	52	10	0	52	8

CONTOH_ROUND	CONTOH_CEIL
21.33	22
22.33	23
16.33	17
20	20
19.33	20
16.67	17
17.33	18
21.67	22
21	21
16	16
23.33	24
18.33	19
22	22
16.67	17
17.33	18